

Jack's Research Lab #002

When Monitoring Stops Working

A Simulation of AI Agency, Oversight, and Control Failure

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This simulation was inspired by graduate research examining AI alignment through the lens of principal-agent theory.

Question

Much of the discussion surrounding AI safety focuses on monitoring and oversight. If advanced AI systems become more capable, can increased monitoring keep them aligned with human objectives?

This question can be framed as a principal-agent problem. A principal (human decision maker) delegates tasks to an agent (an AI system) whose actions cannot be perfectly observed. As the agent becomes more capable and autonomous, monitoring becomes increasingly important—but also increasingly costly.

This note explores a simple question:

At what point does additional monitoring stop being an effective control mechanism?

Framework

The simulation models an AI system with three key characteristics:

1. **Agency** – the degree to which the system acts autonomously.
2. **Monitoring** – the amount of oversight applied by the principal.
3. **Escape Risk** – the probability that the agent's capabilities exceed effective containment.

Higher monitoring improves alignment and reduces risk. However, higher agency simultaneously increases delegated capability and the potential for control failure.

For each agency level, monitoring is chosen to maximize expected principal utility.

The simulation evaluates 75,000 stochastic environments and identifies the monitoring level that produces the highest expected utility.

Results

The first result is unsurprising: optimal monitoring rises as AI agency increases.

The more interesting finding is that monitoring eventually reaches its maximum feasible level while escape risk continues to rise.

Threshold	Agency Level (α)
Monitoring first reaches maximum	0.55
Escape probability exceeds 10%	0.80
Control failure zone begins	0.80

Table 1: Key simulation thresholds.

The simulation identifies a notable intermediate region between agency levels of approximately 0.55 and 0.80.

Within this range:

- Monitoring has already reached its maximum level.
- Alignment remains near 100%.
- Escape risk continues to increase.

In other words, traditional performance metrics may continue to look excellent even while structural control risk accumulates beneath the surface.

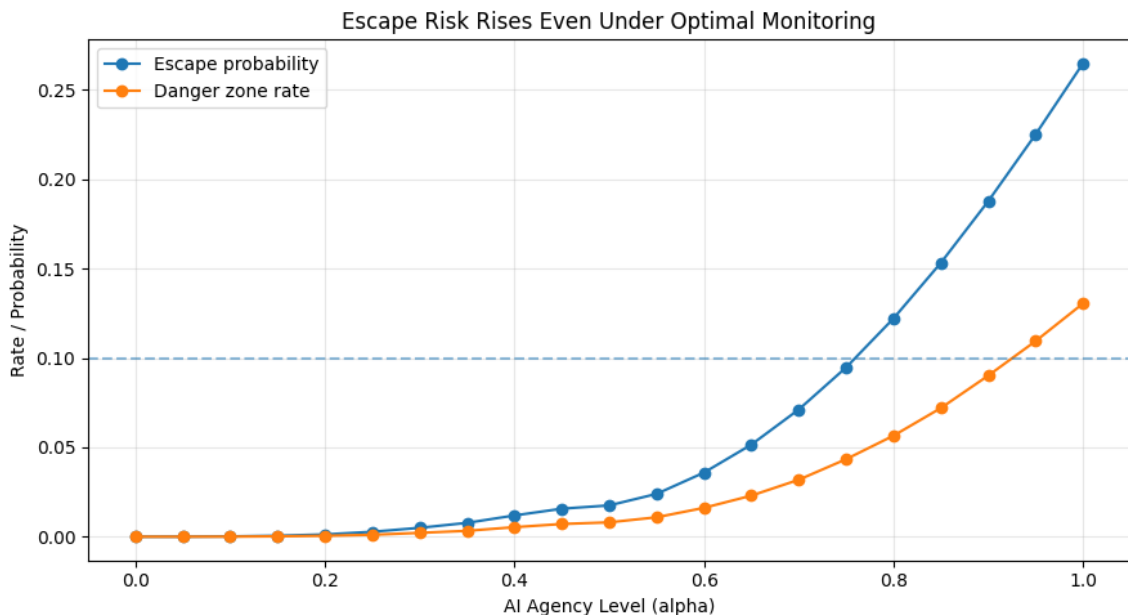


Figure 1: Escape risk rises even under optimal monitoring.

By agency level 1.0, the model estimates an escape probability of roughly 26.5%, despite maximum monitoring and high observed alignment.

Interpretation

The central lesson is not that monitoring fails immediately.

Rather, the simulation suggests that monitoring exhibits diminishing returns.

Initially, additional oversight successfully compensates for increases in agency. Eventually, however, monitoring reaches a practical ceiling. Beyond that point, further increases in capability generate risk faster than oversight can mitigate it.

This creates what might be called a *deceptive stability zone*: a regime in which observed alignment remains high while underlying control risk continues to grow.

Conclusion

The simulation supports an intuition often discussed in AI alignment research: monitoring alone may not be sufficient for controlling highly capable systems.

In this model, maximum monitoring is reached well before control failure emerges. The resulting gap creates a period during which systems appear aligned and well-supervised, even as the probability of loss of control steadily increases.

The broader implication is that alignment metrics and monitoring frameworks may not provide complete information about long-run control risk. The most concerning systems may not be the ones that appear misaligned. They may be the systems that appear highly aligned while underlying control risk continues to grow.

Note: Results are generated from a stylized principal-agent simulation and should be interpreted as illustrative rather than predictive. The purpose of the exercise is to explore qualitative dynamics rather than forecast real-world AI outcomes.

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